

SNMP – Simple Network Management Protocol

1. What is SNMP?

SNMP = Simple Network Management Protocol

SNMP is an **application-layer protocol/framework** used to **monitor and manage network devices**.

It can be used to:

- Monitor the status of routers and switches
- Check interface status
- Monitor traffic
- Monitor CPU and memory usage
- Receive alerts when something goes wrong
- Change certain configuration values remotely

SNMP was originally released in **1988**.

RFCs related to SNMPv1

- **RFC 1065** – Structure and identification of management information
- **RFC 1066** – Management Information Base (MIB)
- **RFC 1067** – Simple Network Management Protocol

Image 1 –



Simple Network Management Protocol

- SNMP is an industry-standard framework and protocol that was originally released in 1988.
RFC 1065 – Structure and identification of management information for TCP/IP-based internets
RFC 1066 – Management information base for network management of TCP/IP-based internets
RFC 1067 – A simple network management protocol
- Don't let the 'Simple' in the name fool you! **SNMPv1**
- SNMP can be used to monitor the status of devices, make configuration changes, etc.
- There are two main types of devices in SNMP:
 - 1) Managed Devices
 - These are the devices being managed using SNMP.
 - For example, network devices like routers and switches.
 - 2) Network Management Station (NMS)
 - The device/devices managing the managed devices.
 - This is the SNMP 'server'.



2. Two Main Types of Devices in SNMP

There are two important components at the device/network level:

1. Managed Device

A **managed device** is the network device being monitored or managed using SNMP.

Examples:

- Router
- Switch
- Firewall
- Access Point
- Server

Examples from the diagram:

R1 = Managed Device

SW1 = Managed Device

A managed device runs an **SNMP Agent**.

2. Network Management Station (NMS)

NMS = Network Management Station

The NMS is the system that manages and monitors the managed devices.

It usually contains:

- SNMP Manager
- SNMP Application

The NMS can:

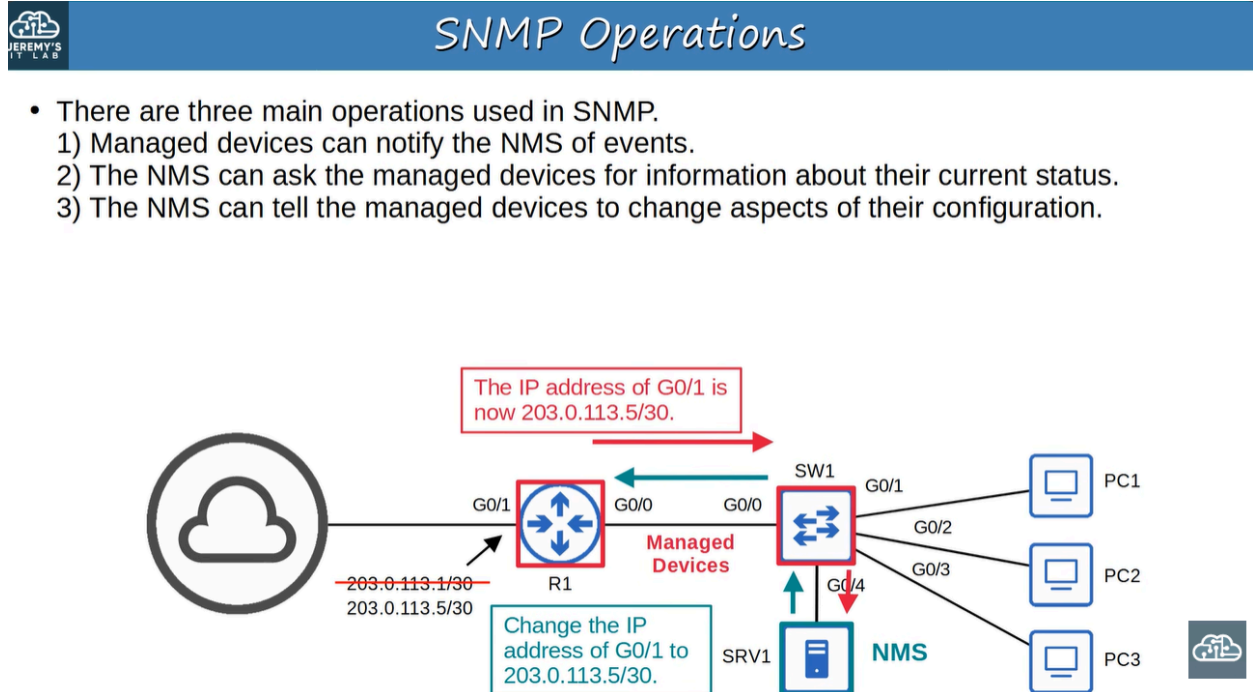
- Request information
- Receive notifications
- Send configuration changes
- Display alerts
- Display statistics
- Display graphs/charts

Simple understanding

NMS

```
|  
|---- SNMP Manager  
|  
|---- SNMP Application
```

Image 2 –



It shows:

NMS

|

+---- SNMP Manager

|

+---- SNMP Application

|

+---- Managed Devices

3. SNMP Operations

There are three main types of operations:

1. Managed Device → NMS

The managed device can **notify the NMS about an event**.

Example:

Router interface goes DOWN

↓
SNMP Agent detects it
↓
Notification sent to NMS

2. NMS → Managed Device

The NMS can **ask the managed device for information**.

Example:

NMS → "What is the status of G0/1?"
↓
Router
↓
"G0/1 is UP"

3. NMS → Managed Device

The NMS can **tell the managed device to change certain configuration information**.

Example:

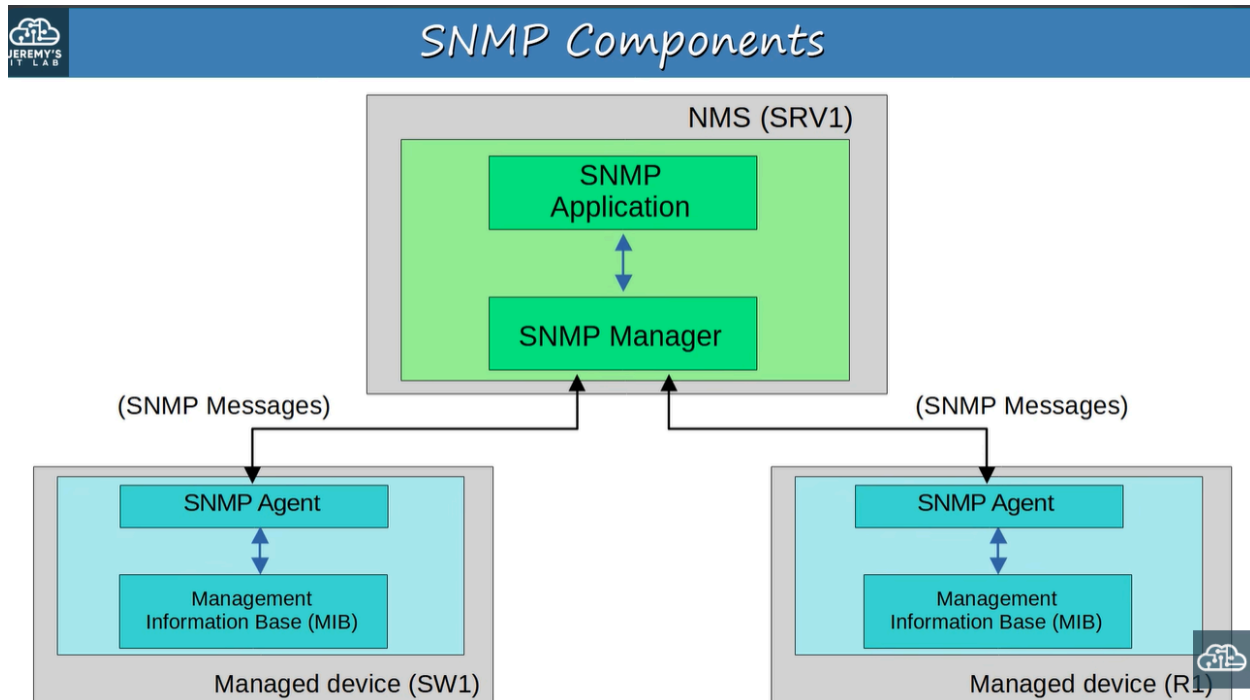
NMS → Change hostname to SW10
↓
Switch
↓
Configuration changed

Important

SNMP does not mean the NMS can automatically execute every possible CLI command.

A device can change only the values that are exposed as **writable SNMP objects/OIDs**.

Image 3 –

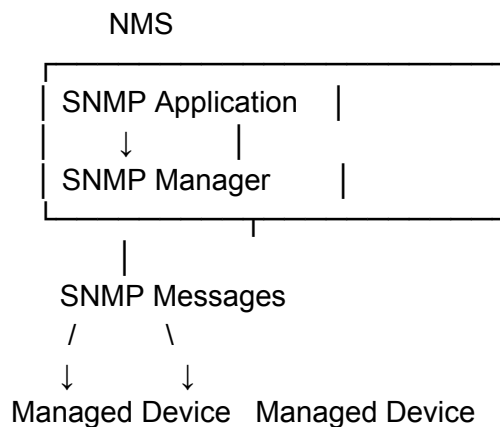


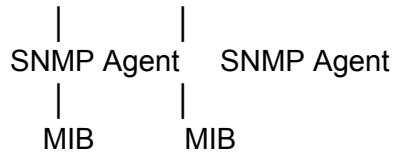
This image clearly shows:

- Device notification
- NMS asking for information
- NMS changing configuration

4. SNMP Components

The major SNMP components are:





The four important terms are:

1. SNMP Manager
 2. SNMP Application
 3. SNMP Agent
 4. MIB
-

5. SNMP Manager

The **SNMP Manager** is software running on the NMS.

It communicates with SNMP Agents running on managed devices.

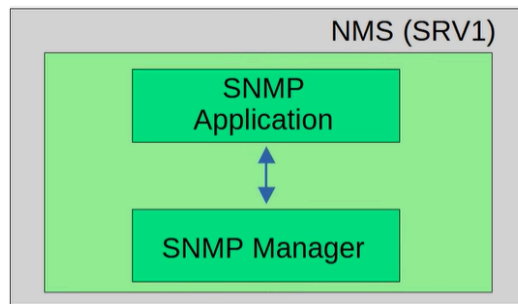
The SNMP Manager can:

- Send requests
- Receive responses
- Receive notifications
- Send configuration changes

Easy way to remember:

Manager = controls/requests/receives

Image 4 –



- The **SNMP Manager** is the software on the NMS that interacts with the managed devices.
→ It receives notifications, sends requests for information, sends configuration changes, etc.
- The **SNMP Application** provides an interface for the network admin to interact with.
→ Displays alerts, statistics, charts, etc.

The image shows:

NMS



6. SNMP Application

The **SNMP Application** provides the interface used by the network administrator.

It can display:

- Alerts
- Statistics
- Charts
- Device status
- Monitoring information

Important difference

SNMP Manager



Communicates with devices

SNMP Application



Provides interface for administrator

The administrator normally interacts with the **SNMP application**, while the **SNMP Manager** handles SNMP communication.

Image 5 – Where to paste

Paste Image 5 after the SNMP Agent and MIB section below, or immediately after Image 4 if you want the NMS components together.

7. SNMP Agent

The **SNMP Agent** is SNMP software running on the managed device.

Examples:

Router → SNMP Agent

Switch → SNMP Agent

The Agent:

- Receives SNMP requests from the Manager
- Provides information to the Manager
- Sends notifications to the Manager
- Can process configuration changes requested through SNMP

Easy way to remember

Agent = lives on the managed device.

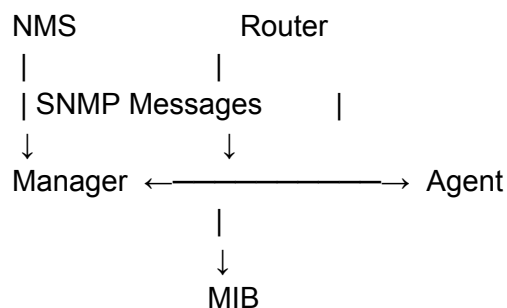


Image 6

SNMP Components

- The **SNMP Agent** is the SNMP software running on the managed devices that interacts with the SNMP Manager on the NMS.
→ It sends notifications to/receives messages from the NMS.
- The **Management Information Base (MIB)** is the structure that contains the variables that are managed by SNMP.
→ Each variable is identified with an Object ID (OID)
→ Example variables: Interface status, traffic throughput, CPU usage, temperature, etc.

< - 5



8. MIB – Management Information Base

MIB = Management Information Base

The MIB is the structure/collection that defines the **managed objects/variables available through SNMP**.

Examples:

- Interface status
- Interface traffic
- CPU usage
- Temperature
- Device name
- Device uptime

Each managed variable is identified by an **OID**.

Important clarification

Think of:

MIB = organized collection/definition of manageable information

OID = unique identifier for a particular object

The actual current value is provided by the SNMP Agent.

Example:

MIB

- |
- +— Interface status
- +— CPU usage
- +— System name
- +— Uptime

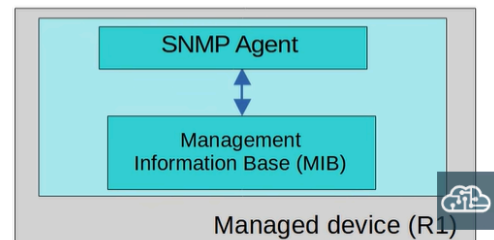
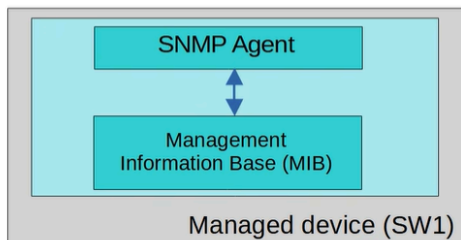
Image 7 – Where to paste



SNMP Components

- The **SNMP Agent** is the SNMP software running on the managed devices that interacts with the SNMP Manager on the NMS.
→ It sends notifications to/receives messages from the NMS.
- The **Management Information Base (MIB)** is the structure that contains the variables that are managed by SNMP.
→ Each variable is identified with an Object ID (OID)
→ Example variables: Interface status, traffic throughput, CPU usage, temperature, etc.

< - 5



It visually shows:

SNMP Agent



MIB

9. OID – Object Identifier

OID = Object Identifier

An OID uniquely identifies a particular SNMP object.

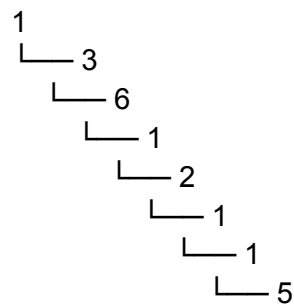
OIDs are organized in a **hierarchical tree structure**.

Example:

1.3.6.1.2.1.1.5

This identifies the **sysName** object in the example shown in the slide.

The hierarchy can be thought of as:



In the example:

OID: 1.3.6.1.2.1.1.5

Value: SW1

So the NMS asks the Agent for that OID, and the Agent returns its current value.

Very important

MIB = What objects are available

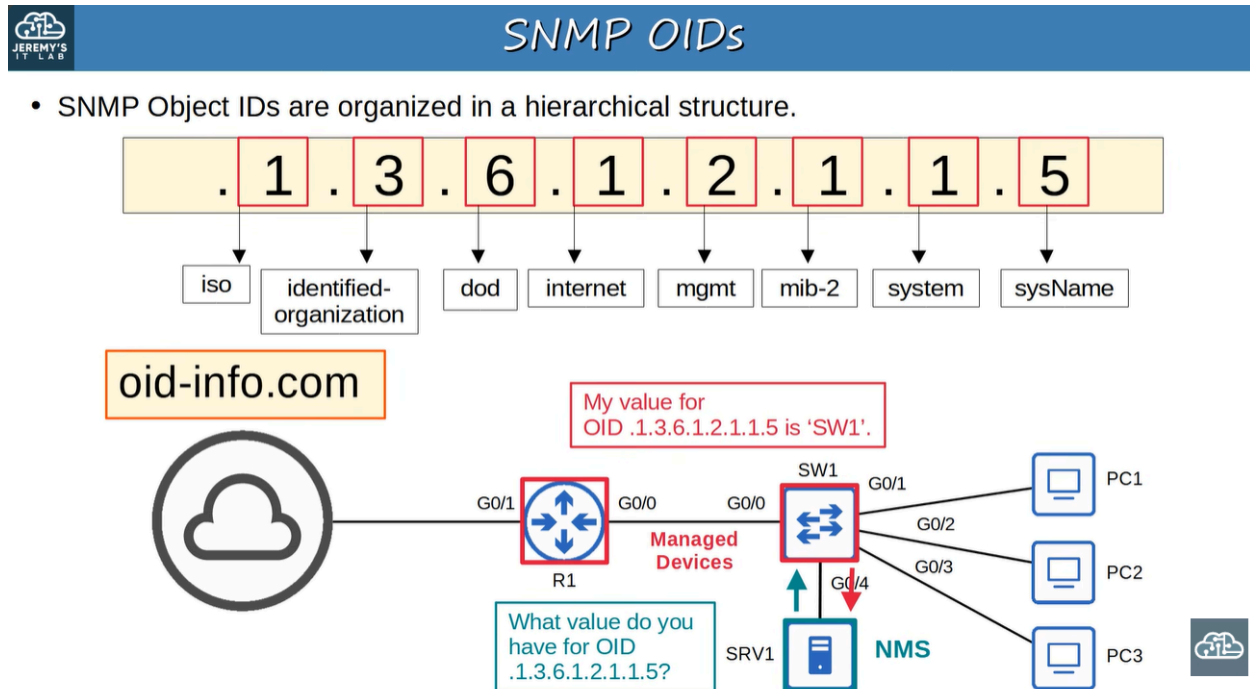
OID = Which exact object are we talking about?

Value = Current value of that object

Example:

OID → sysName
Value → SW1

Image 8 – Where to paste



This is the best image for remembering the OID hierarchy.

10. SNMP Versions

Three major versions are commonly discussed:

SNMPv1

- Original version
- Basic SNMP version
- Uses community strings
- No encryption

SNMPv2c

Advantages over v1 include:

- More efficient operations
- Supports **GetBulk**
- Can retrieve larger amounts of information efficiently
- Uses community strings
- No encryption

The "c" refers to **community-based security**.

Important

SNMPv2c community strings are **not encryption**.

They can be sent in plain text.

SNMPv3

SNMPv3 provides stronger security.

It supports:

- Authentication
- Integrity
- Encryption/privacy

Therefore:

SNMPv1 → Basic

SNMPv2c → More efficient

SNMPv3 → Strong security

Image 9



SNMP Versions

- Many versions of SNMP have been proposed/developed, however only three major versions have achieved wide-spread use:
- **SNMPv1**
 - The original version of SNMP.
- **SNMPv2c**
 - Allows the NMS to retrieve large amounts of information in a single request, so it is more efficient. ☺
 - 'c' refers to the 'community strings' used as passwords in SNMPv1, removed from SNMPv2, and then added back for SNMPv2c.
- **SNMPv3**
 - A much more secure version of SNMP that supports strong **encryption** and **authentication**. Whenever possible, this version should be used!



11. SNMP Message Types



SNMP Messages

Message Class	Description	Messages
Read	Messages sent by the NMS to read information from the managed devices . (ie. What's your current CPU usage %?)	<i>Get</i> <i>GetNext</i> <i>GetBulk</i>
Write	Messages sent by the NMS to change information on the managed devices . (ie. change an IP address)	<i>Set</i>
Notification	Messages sent by the managed devices to alert the NMS of a particular event. (ie. interface going down)	<i>Trap</i> <i>Inform</i>
Response	Messages sent in response to a previous message/request.	<i>Response</i>



SNMP messages can be divided into four categories:

Category	Messages
----------	----------

Read	Get, GetNext, GetBulk
------	-----------------------

Write	Set
-------	-----

Notification Trap, Inform

Response Response

12. GET

GET is a request sent by the SNMP Manager to the Agent to retrieve the value of an OID.

Example:

NMS

|

| GET: What is G0/1 status?

↓

Router

|

| RESPONSE: G0/1 is UP

↓

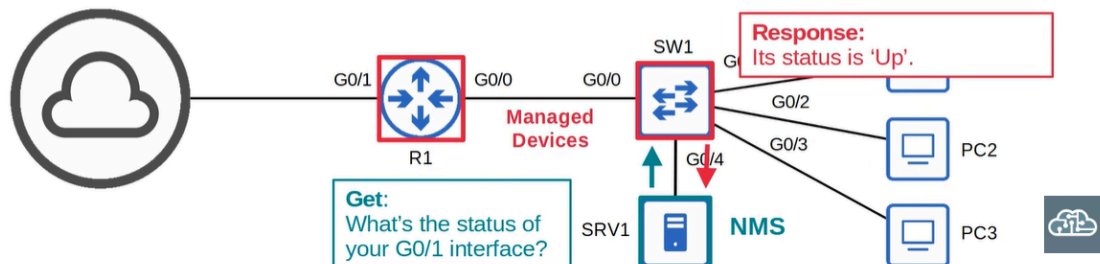
NMS

Easy memory:

GET = "Give me this information."

Image 10

- **Get**
→ A request sent from the manager to the agent to retrieve the value of a variable (OID), or multiple variables. The agent will send a *Response* message with the current value of each variable.
- **GetNext**
→ A request sent from the manager to the agent to discover the available variables in the MIB.
- **GetBulk**
→ A more efficient version of the **GetNext** message (introduced in SNMPv2).



13. GetNext

GetNext is used to discover/retrieve the next available variable in the MIB hierarchy.

It is useful when the Manager does not already know the exact next object it wants.

Simple idea

```
OID 1
↓
GetNext
↓
OID 2
↓
GetNext
↓
OID 3
```

14. GetBulk

GetBulk is an efficient version of retrieving multiple values.

It was introduced with SNMPv2.

Instead of sending many individual requests, the NMS can request a larger amount of information efficiently.

Easy memory

GET → Get a value

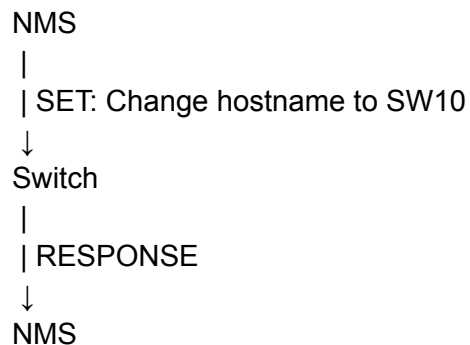
GetNext → Get the next object

GetBulk → Get many objects efficiently

15. SET

SET is a request sent by the SNMP Manager to change the value of an SNMP-managed object.

Example:



Important

SET is used only for **writable SNMP objects**.

It does not mean every router/switch configuration can be changed through SNMP.

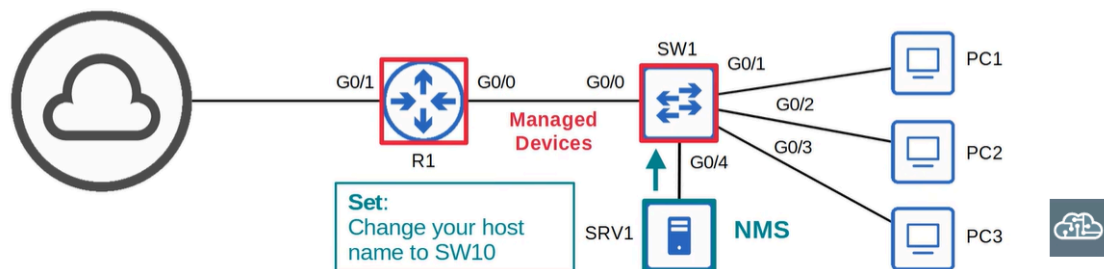
Easy memory:

SET = "Change this value."

SNMP 'Write' Messages

- **Set**

→ A request sent from the manager to the agent to change the value of one or more variables. The agent will send a *Response* message with the new values.



16. TRAP

A **Trap** is an unsolicited notification sent from the SNMP Agent to the SNMP Manager.

The Agent sends it when an important event occurs.

Example:

Interface G0/0 goes DOWN

```
graph TD; A[ ] --> B[SNMP Agent]; B --> C[TRAP]; C --> D[SNMP Manager];
```

Example message:

"My G0/0 interface went down!"

Important characteristic

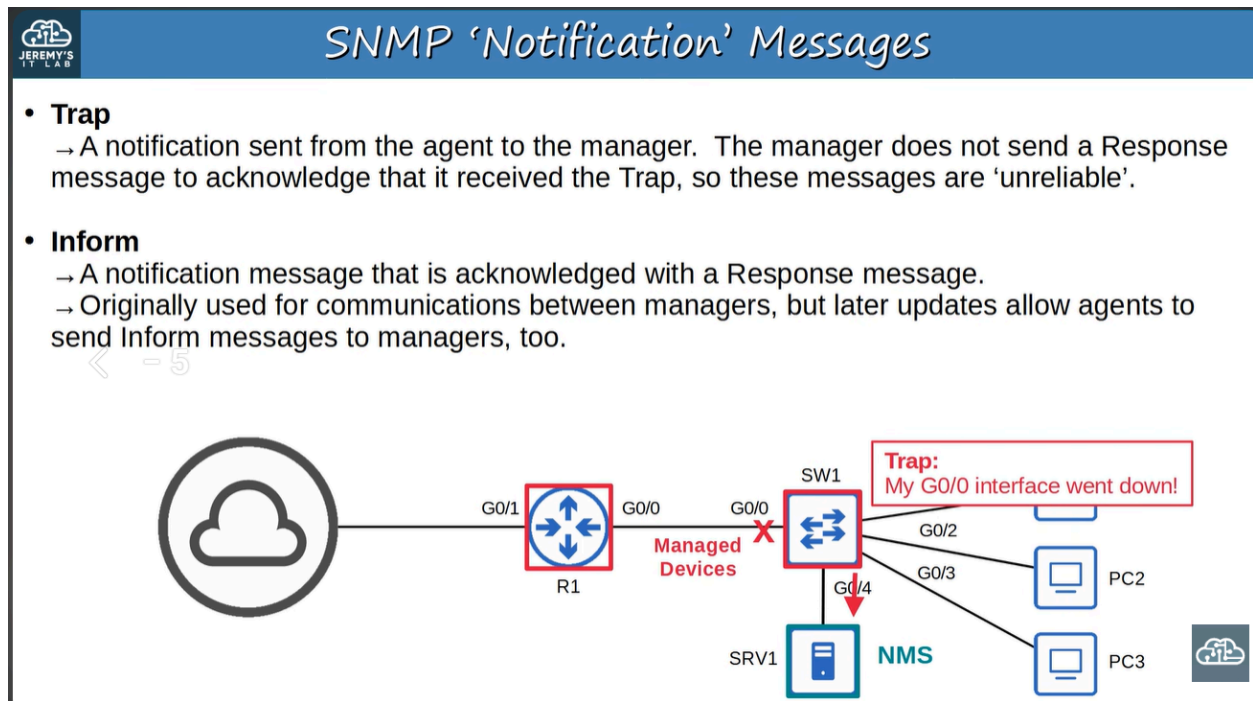
A traditional SNMP **Trap** **does not require an acknowledgement/response** from the Manager.

Therefore, it is considered **unacknowledged/unreliable**.

Easy memory:

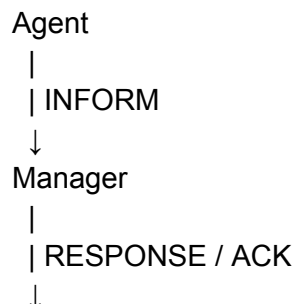
TRAP = "Hey Manager! Something happened!"

Image 12



17. INFORM

Inform is similar to a Trap, but it expects an acknowledgement.



Agent

Therefore:

TRAP

→ Notification

→ No acknowledgement

INFORM

→ Notification

→ Acknowledgement/Response

Important difference

Feature	Trap	Inform
Notification	Yes	Yes
Acknowledgement	No	Yes
Reliability	Lower	Higher
Response expected	No	Yes

Easy memory

Trap = Tell

Inform = Tell + Confirm

18. Response

A **Response** is sent in response to a previous SNMP request.

Example:

Manager → GET → Agent

Agent → RESPONSE → Manager

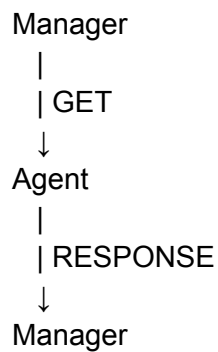
Example:

GET: What is G0/1 status?

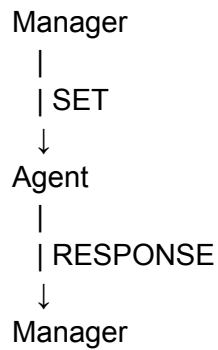
RESPONSE: UP

19. Complete SNMP Message Flow

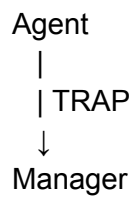
GET



SET

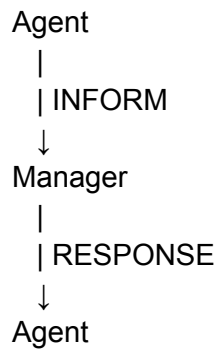


TRAP



No acknowledgement is required.

INFORM



20. SNMP Ports

SNMP commonly uses **UDP**.

UDP 161

Used for normal SNMP request/response communication with the Agent.

Manager → Agent
UDP 161

Examples:

- GET
- GETNEXT
- GETBULK
- SET
- Response

UDP 162

Used by the SNMP Manager/NMS to receive **notifications such as Trap and Inform**.

Agent → Manager
UDP 162

Important clarification about the slide

The slide says:

SNMP Agent = UDP 161

SNMP Manager = UDP 162

This is a simplified way of remembering the **destination ports**.

It does NOT mean:

"The SNMP Manager always communicates using UDP 162."

Correct understanding:

Normal SNMP requests → Agent UDP 161

Trap/Inform → Manager UDP 162

Image 13 – Where to paste

Paste Image 13 immediately after the SNMP ports explanation.

21. SNMPv2c Configuration

Example Cisco configuration:

```
R1(config)# snmp-server contact jeremy@jeremysitlab.com
```

```
R1(config)# snmp-server location Jeremy's House
```

These provide optional information about:

- Contact person
 - Device location
-

Configure Community Strings

```
R1(config)# snmp-server community Jeremy1 ro
```

```
R1(config)# snmp-server community Jeremy2 rw
```

ro

Read Only

The NMS can retrieve information but cannot use SET operations through that community.

ro → Read only

rw

Read/Write

The NMS can retrieve information and perform SET operations allowed by the device.

rw → Read + Write

Easy memory

ro → GET

rw → GET + SET

22. Configure the NMS

Example:

```
R1(config)# snmp-server host 192.168.1.1 version 2c Jeremy1
```

This tells R1:

NMS IP = 192.168.1.1

SNMP version = 2c

Community = Jeremy1

The router now knows where to send SNMP notifications.



SNMPv2c Configuration

```
R1(config)#snmp-server contact jeremy@jeremysitlab.com
R1(config)#snmp-server location Jeremy's House
```

Optional information

```
R1(config)#snmp-server community Jeremy1 ro
R1(config)#snmp-server community Jeremy2 rw
```

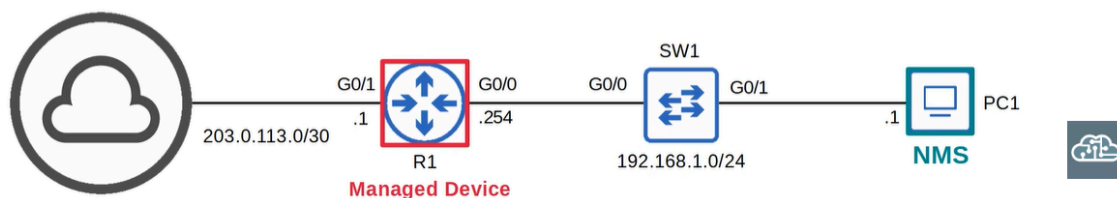
Configure the SNMP *community strings* (passwords)
ro = read only = no Set messages
rw = read/write = can use Set messages

```
R1(config)#snmp-server host 192.168.1.1 version 2c Jeremy1
```

Specify the NMS, version, and community

```
R1(config)#snmp-server enable traps snmp linkdown linkup
R1(config)#snmp-server enable traps config
```

Configure the Trap types to send to the NMS



23. Enable SNMP Traps

Example:

```
R1(config)# snmp-server enable traps snmp linkdown linkup
R1(config)# snmp-server enable traps config
```

This enables selected types of SNMP notifications.

For example:

linkdown



Interface goes DOWN



SNMP Trap



NMS

It shows the actual Cisco commands and topology, so it is useful for practical/exam revision.

24. Wireshark and SNMPv2c

In Wireshark, an SNMP packet can show information such as:

Version: v2c

Community: Jeremy1

Destination Port: 162

This demonstrates an important security issue:

SNMPv1 and SNMPv2c do NOT provide encryption.

The:

- Community string
- SNMP message contents

can be transmitted in plain text.

Therefore, someone capable of capturing the traffic could potentially read the community string and message contents.

Example:

SNMPv2c packet

↓

Community: Jeremy1

↓

Plain text

SNMPv3

SNMPv3 provides stronger security features including authentication and privacy/encryption.

Important exam point

SNMPv1 → No encryption

SNMPv2c → No encryption

SNMPv3 → Supports authentication + encryption/privacy

Image 15 –



Wireshark Capture

No.	Time	Source	Destination	Protocol	Length	Info
209	13:55:21.662570	192.168.1.254	192.168.1.1	SNMP	221	snmpV2-trap 1.3.6.1.2.1.

> Frame 209: 221 bytes on wire (1768 bits), 221 bytes captured (1768 bits) on interface -, id 0

> Ethernet II, Src: 0c:1c:1a:87:fb:00 (0c:1c:1a:87:fb:00), Dst: 0c:1c:1a:50:80:01 (0c:1c:1a:50:80:01)

> Internet Protocol Version 4, Src: 192.168.1.254, Dst: 192.168.1.1

> User Datagram Protocol, Src Port: 65385, Dst Port: 162

▼ Simple Network Management Protocol

version: v2c (1)

community: Jeremy1

▼ data: snmpV2-trap (7)

▼ snmpV2-trap

request-id: 14

error-status: noError (0)

error-index: 0

▼ variable-bindings: 6 items

> 1.3.6.1.2.1.1.3.0: 104924

> 1.3.6.1.6.3.1.1.4.1.0: 1.3.6.1.6.3.1.1.5.3 (iso.3.6.1.6.3.1.1.5.3)

> 1.3.6.1.2.1.2.2.1.1.2: 2

> 1.3.6.1.2.1.2.2.1.2.2: 4769676162697445746865726e6574302f31

> 1.3.6.1.2.1.2.2.1.3.2: 6

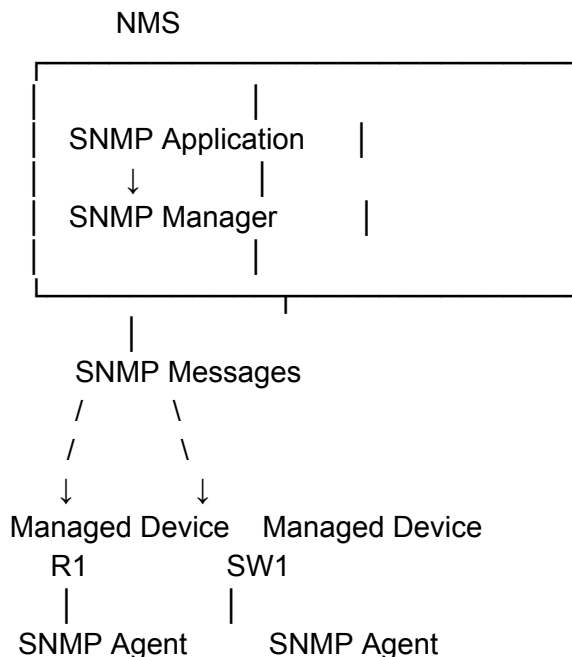
> 1.3.6.1.4.1.9.2.2.1.1.20.2: 61646d696e6973747261746976656c7920646f776e

In SNMPv1 and SNMPv2c, there is no encryption. The community and message contents are sent in plain-text. This is not secure, as the packets can easily be captured and read.



25. Complete SNMP Architecture

Remember this structure:



|
MIB

|
MIB

26. SNMP – One-Line Meaning of Every Term

Term	Meaning
SNMP	Protocol/framework for monitoring and managing devices
NMS	System used to monitor/manage network devices
SNMP Manager	Software on NMS that communicates with Agents
SNMP Application	Interface used by administrator
Managed Device	Device being monitored/managed
SNMP Agent	SNMP software running on managed device
MIB	Collection/structure of manageable objects
OID	Unique identifier of a managed object
GET	Read a specific value
GetNext	Retrieve the next object
GetBulk	Retrieve many values efficiently
SET	Change a writable value
Response	Reply to a request
Trap	Notification without acknowledgement
Inform	Notification with acknowledgement
UDP 161	Normal SNMP request/response port
UDP 162	Trap/Inform destination port
SNMPv1	Original/basic version

SNMPv2c	More efficient, community-based, no encryption
SNMPv3	Security-focused version

27. Most Important Differences

GET vs SET

GET → Read information

SET → Change information

TRAP vs INFORM

TRAP

→ Agent informs Manager

→ No acknowledgement

INFORM

→ Agent informs Manager

→ Manager sends Response/acknowledgement

MIB vs OID

MIB

→ Collection/structure of manageable objects

OID

→ Unique identifier for one particular object

Example:

MIB

└─ System

└─ sysName

└─ OID = 1.3.6.1.2.1.1.5

↓
Value = SW1

Manager vs Agent

SNMP Manager

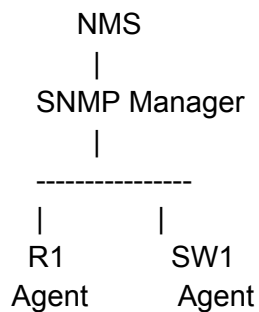
- Runs on NMS
- Sends requests
- Receives responses
- Receives notifications

SNMP Agent

- Runs on managed device
 - Receives requests
 - Sends responses
 - Sends notifications
-

28. Easy Real-World Example

Imagine a company has:



The administrator wants to know whether R1's interface is working.

Step 1

Manager sends:

GET → Interface status OID

Step 2

R1's Agent checks the corresponding value.

Step 3

Agent sends:

RESPONSE → UP

Now suppose the interface suddenly goes down.

Step 4

The Agent sends:

TRAP → Interface DOWN

The NMS receives the notification and displays an alert.

29. Exam-Focused Summary

SNMP

Simple Network Management Protocol

Used to **monitor and manage network devices**.

Components

NMS

- └─ SNMP Application
- └─ SNMP Manager

Managed Device

- └─ SNMP Agent
- └─ MIB

Operations

GET → Read
GETNEXT → Next object
GETBULK → Many objects
SET → Write/change
TRAP → Notification, no ACK
INFORM → Notification + ACK
RESPONSE → Reply

Ports

UDP 161 → Normal SNMP communication
UDP 162 → Trap/Inform notifications

Versions

SNMPv1 → Original
SNMPv2c → Efficient + community string + no encryption
SNMPv3 → Authentication + privacy/encryption

Most important memory line

Manager asks, Agent answers. Agent can also notify Manager. MIB contains manageable objects, and OID identifies each object.

30. Image Placement – Final Order

Use the screenshots in this order in your notes:

Image	Place after
1	SNMP introduction
2	Managed Device & NMS
3	Three SNMP operations
4	SNMP Manager / NMS components
5	SNMP Manager & Application explanation

- 6 SNMP Agent explanation
- 7 MIB explanation
- 8 OID explanation
- 9 SNMP versions
- 10 GET / Read messages
- 11 SET / Write messages
- 12 TRAP / INFORM
- 13 UDP 161 / UDP 162
- 14 SNMPv2c configuration + Wireshark/security — **place the configuration screenshot first and Wireshark screenshot immediately after it**

Recommended final note structure

SNMP Introduction



Managed Device & NMS



SNMP Operations



SNMP Components



Manager



Agent



MIB



OID



SNMP Versions



SNMP Messages



GET / GetNext / GetBulk



SET



TRAP / INFORM

↓
UDP 161 / 162
↓
SNMPv2c Configuration
↓
Wireshark & Security
↓
Exam Summary

SNMP Agent = UDP 161
SNMP Manager = UDP 162